

Digital Image Processing Project — Phase I Report

Blood Cell Anomaly Detection Using Advanced Image Preprocessing Techniques

Spring Term 2026 | Group 1 — Domain-Specific Project

[Ziad Elsayed Fouad] | [320230025] | [E-JUST]

[Mohamed Hussien Kamal] | [320230008] | [E-JUST]

[Mohamed Adham Elhadidi] | [320230001] | [E-JUST]

Dataset: Blood Cell Anomaly Detection 2025 — Kaggle

1. Abstract

Blood cell anomaly detection is a critical task in clinical hematology, enabling the early diagnosis of life-threatening conditions including anemia, leukemia, sickle-cell disease, and various bacterial or viral infections. Manual microscopic examination of peripheral blood smears is the traditional diagnostic method; however, this approach is labor-intensive, operator-dependent, and susceptible to inter-observer variability. Automated image processing systems offer a robust, reproducible, and scalable alternative that can significantly accelerate the diagnostic workflow.

This project addresses the challenge of automated anomaly detection in blood cell microscopy images using the publicly available Blood Cell Anomaly Detection 2025 dataset sourced from Kaggle. The dataset comprises high-resolution microscopy images spanning multiple cell categories including normal and anomalous red blood cells (RBCs), white blood cells (WBCs), and platelets captured under standardized Giemsa staining conditions.

Phase I of this project focuses exclusively on image preprocessing and enhancement — a foundational step that directly governs the accuracy of any subsequent segmentation or classification algorithm. The proposed preprocessing pipeline consists of three sequential stages. First, Contrast Limited Adaptive Histogram Equalization (CLAHE) is applied to the LAB color space L-channel to improve the local contrast of cell nuclei and cytoplasm boundaries without introducing color distortion. Second, a Median filter is employed in the spatial domain to suppress salt-and-pepper noise and microscopic artifacts arising from staining inconsistencies. Third, a Butterworth High-Pass Filter is constructed and applied in the frequency domain using the Fast Fourier Transform (FFT) to selectively amplify high-frequency structural components such as cell membrane edges and nuclear chromatin patterns, which are clinically significant morphological markers.

Qualitative evaluation of the pipeline demonstrates marked improvements in cell boundary delineation, nucleus-cytoplasm contrast, and suppression of background artifacts, thereby providing a high-quality input for Phase II segmentation and classification stages. The entire pipeline is implemented in Python using OpenCV and NumPy, ensuring reproducibility and ease of integration into a larger machine-learning workflow.

2. Problem Definition

Hematological disorders are among the leading causes of morbidity and mortality worldwide. Accurate identification of morphological anomalies in blood cells — such as sickle-shaped RBCs, blasts in leukemia, or hyper segmented neutrophils — is essential for timely clinical intervention. The current gold-standard diagnostic procedure, the Complete Blood Count (CBC) combined with manual peripheral blood smear examination, suffers from the following critical limitations:

- Time and resource intensity: A single smear analysis requires 15–30 minutes of an expert hematologist's time, creating bottlenecks in high-volume clinical labs.
- Subjectivity and variability: Inter-observer agreement rates for anomaly classification can be as low as 70–80%, especially for rare cell types.
- Image quality degradation: Raw microscopy images are frequently degraded by uneven illumination, staining artifacts, out-of-focus regions, and sensor noise, all of which impair both human and machine analysis.
- Scalability: Manual workflows cannot scale to the demands of large epidemiological studies or resource-limited healthcare settings.

Automated image analysis systems must therefore address these limitations. However, before any machine learning model can reliably detect anomalies, the input images must be preprocessed to reduce noise, normalize contrast, and emphasize discriminative morphological features. Without robust preprocessing, downstream classifiers are exposed to irrelevant variability in illumination and noise, leading to degraded precision and recall. This project directly tackles this preprocessing challenge as its Phase I contribution.

3. Objectives

- Develop a reproducible, automated preprocessing pipeline for blood cell microscopy images.
- Enhance cell nucleus and membrane contrast using CLAHE in the LAB color space.
- Remove microscopic staining artifacts and sensor noise through median filtering.
- Apply a frequency-domain Butterworth High-Pass Filter via FFT to accentuate structural edges.
- Provide qualitative visual evidence of preprocessing improvements at each stage.
- Prepare a clean, high-quality image set as input for Phase II segmentation and classification.

4. Dataset Description

The Blood Cell Anomaly Detection 2025 dataset is a publicly available benchmark hosted on Kaggle (<https://www.kaggle.com/datasets/alitaqishah/blood-cell-anomaly-detection-2025>). It was curated to support research in automated hematological image analysis and contains microscopy images of peripheral blood smears prepared with standard Giemsa staining protocol.

Property	Details
Image Format	JPEG / PNG, RGB color
Image Resolution	High-resolution microscopy (varies by slide)
Cell Types Covered	Red Blood Cells (RBCs), White Blood Cells (WBCs), Platelets
Classes	Normal cells and anomalous/diseased cells
Staining Protocol	Giemsa stain — standard for blood smear analysis
Source	Kaggle — alitaqishah/blood-cell-anomaly-detection-2025
Year	2025

The images capture cells at different magnifications and represent a range of morphological variations. Common anomalies present in the dataset include sickle cells, target cells, elliptocytes, and blast cells. The diversity of cell appearances and background conditions makes robust preprocessing essential.

5. Methodology

The preprocessing pipeline is implemented in Python 3.x using OpenCV and NumPy. Three sequential stages are applied to each image, as detailed below. The complete annotated source code is submitted alongside this report.

5.1 Stage 1 — CLAHE (Contrast Limited Adaptive Histogram Equalization)

Standard global histogram equalization redistributes pixel intensities uniformly across the entire image, which is poorly suited to microscopy images where cells occupy only a fraction of the frame and background brightness varies locally. CLAHE overcomes this by partitioning the image into small non-overlapping tiles and performing equalization within each tile independently, subject to a contrast clip limit that prevents over-amplification of noise.

Implementation details:

- The image is converted from BGR to LAB color space. CLAHE is applied exclusively to the L (luminance) channel, preserving the hue and saturation information encoded in A and B channels, thus avoiding color distortion.
- Clip limit: 2.0 — balances contrast enhancement against noise amplification.
- Tile grid: 8 × 8 — fine enough for local contrast normalization without over-segmenting the image.

After enhancement, the image is converted back to BGR. The result shows markedly improved visibility of cell nuclei, cytoplasm boundaries, and internal granular structures.

5.2 Stage 2 — Median Filter (Spatial-Domain Noise Removal)

Microscopy images are commonly affected by salt-and-pepper noise arising from staining inconsistencies, dust particles on slides, and CCD sensor defects. Unlike Gaussian filtering, the median filter replaces each pixel with the median value of its neighborhood, making it non-linear and highly effective at removing impulse noise while preserving sharp edges — a property of paramount importance for cell boundary detection.

Implementation details:

- Kernel size: 3×3 — preserves fine cell structures while effectively suppressing isolated noise pixels.
- Applied to the full BGR image (all three channels) after CLAHE enhancement.
- OpenCV's `cv2.medianBlur()` is used for computational efficiency.

5.3 Stage 3 — Butterworth High-Pass Filter (Frequency Domain)

The Butterworth High-Pass Filter (BWHPF) is applied in the frequency domain using the 2-D Fast Fourier Transform (FFT). This stage selectively attenuates low-frequency background variation (slow illumination gradients) while preserving and amplifying high-frequency components corresponding to cell membrane edges and intra-nuclear chromatin texture — critical diagnostic features.

The Butterworth HPF transfer function is defined as:

$$H(u, v) = 1 - 1 / [1 + (D(u, v) / D_0)^{2n}]$$

where $D(u, v)$ is the Euclidean distance from the DC component, D_0 is the cut-off frequency, and n is the filter order.

Implementation details:

- Filter order $n = 2$ — provides a smooth roll-off without ringing artifacts (Gibbs phenomenon).
- Cut-off ratio = $0.10 \times \min(\text{rows}, \text{cols}) / 2$ — retains fine structural details while removing low-frequency background variation.
- Procedure: (1) Compute 2-D FFT; (2) shift DC component to image centre; (3) multiply by HPF mask; (4) apply inverse FFT; (5) normalise to $[0, 255]$.

6. Preprocessing Pipeline Summary

Step	Technique	Domain	Purpose
1	CLAHE (L-channel, LAB space)	Spatial	Local contrast enhancement of cell nuclei
2	Median Filter (3×3 kernel)	Spatial	Salt-and-pepper / artifact noise removal
3	Butterworth HPF ($n=2$, $D_0=10\%$)	Frequency (FFT)	Edge / high-frequency structure amplification

7. Results and Interpretation

The following qualitative observations were made by visually comparing each preprocessing stage against the original raw image:

7.1 After CLAHE

The CLAHE-enhanced images demonstrate substantially improved nucleus-cytoplasm contrast. Cell nuclei, which were previously indistinct against the pink-stained cytoplasm background, become clearly delineated with enhanced internal chromatin structure visible. Local contrast improvements are consistent across different regions of the image without introducing global over-brightening or color shifts. The LAB-space approach ensures that the characteristic Giemsa staining colors are fully preserved.

7.2 After Median Filtering

The median-filtered images show a clean suppression of isolated bright and dark noise pixels that were visible in the CLAHE output. Background regions between cells appear smoother and more uniform. Importantly, cell membrane boundaries and nuclear perimeters remain sharp, confirming the edge-preserving property of median filtering. Staining artifact speckles — a common issue in Giemsa-stained smears — are effectively removed.

7.3 After Butterworth HPF (Frequency Domain)

The frequency-domain filtered output highlights cell boundary contours and intra-cellular texture with high fidelity. The slow background illumination gradient — which dominated the low-frequency components of the FFT magnitude spectrum — is effectively suppressed. High-frequency components corresponding to cell edges are amplified, producing an image well-suited for edge-based segmentation algorithms in Phase II. The FFT magnitude spectrum confirms the attenuation of the central DC region and the preservation of peripheral high-frequency energy.

Overall, the three-stage pipeline produces output images with superior cell boundary clarity, reduced background clutter, and enhanced structural feature visibility compared to raw inputs, providing a strong foundation for Phase II segmentation and classification tasks.

8. Tools and Libraries

Tool / Library	Version	Purpose
Python	3.10+	Primary programming language
OpenCV (cv2)	4.x	Image I/O, CLAHE, median filter
NumPy	1.x	FFT, array operations, filter construction
Matplotlib	3.x	Visualization of pipeline stages and spectra

9. Conclusion

Phase I of the Blood Cell Anomaly Detection project has successfully established a robust, automated image preprocessing pipeline. The sequential application of CLAHE for contrast enhancement, Median filtering for noise removal, and a Butterworth High-Pass Filter in the frequency domain collectively deliver significant qualitative improvements in blood cell image quality. The pipeline is fully implemented in Python and is ready to serve as the preprocessing stage for Phase II, where advanced segmentation, CNN-based feature extraction, and machine learning classification will be applied to achieve automated anomaly detection.

References

- [1] Shah, A. T. (2025). Blood Cell Anomaly Detection 2025 [Dataset]. Kaggle. <https://www.kaggle.com/datasets/alitaqishah/blood-cell-anomaly-detection-2025>
- [2] Gonzalez, R. C., & Woods, R. E. (2018). Digital Image Processing (4th ed.). Pearson Education.
- [3] Zuiderveld, K. (1994). Contrast Limited Adaptive Histogram Equalization. In Graphics Gems IV. Academic Press Professional.
- [4] Huang, T. S., Yang, G. J., & Tang, G. Y. (1979). A fast two-dimensional median filtering algorithm. IEEE Transactions on Acoustics, Speech, and Signal Processing, 27(1), 13–18.
- [5] Butterworth, S. (1930). On the theory of filter amplifiers. Wireless Engineer, 7, 536–541.
- [6] Bradski, G., & Kaehler, A. (2008). Learning OpenCV. O'Reilly Media.